

### 3. Project Design Phase

#### 3.1. System Architecture

The Credit Card Approval Prediction System follows a machine learning-based architecture consisting of the following components:

1. User Interface

Collects customer details such as income, age, employment status, and credit history.

2. Data Preprocessing Layer

Cleans the input data, handles missing values, encodes categorical data, and prepares features.

3. Machine Learning Model

The trained classification model analyzes processed data and predicts credit card approval status.

4. Prediction Module

Generates the final output as Approved or Rejected.

5. Output Interface

Displays the prediction result to the user.

Flow:

User Input → Data Preprocessing → ML Model → Prediction → Result Display

#### 3.2. Data Flow Diagram (DFD - Level 1)

Data Flow Diagram (DFD) - Credit Card Approval Prediction System

Level 1 DFD:

User Input



Data Collection Module



Data Preprocessing

(Handling missing values, encoding, feature scaling)



Machine Learning Model

(Trained prediction algorithm)



Prediction Module



Approval/Rejection Result



User Data Storage

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